(DEMO) From Natural Language to Live 5G API Calls: A CAPIF-Compliant, Multi-LLM Intent Orchestration Platform

Türkan Doğa Durak*, Engin Zeydan‡, Luis Blanco‡

*Universitat Politècnica de Catalunya (UPC), Barcelona, Spain.
‡Centre Tecnològic de Telecomunicacions de Catalunya (CTTC/CERCA), Castelldefels, Spain, 08860.
Email: turkan.doga.durak@estudiantat.upc.edu, {ezeydan, luis.blanco}@cttc.es

***Abstract*—We demonstrate an open-source platform that converts free-text operator intents into live CAMARA API calls. A two-phase large-language-model (LLM) planner selects the CAMARA service and builds a schema-grounded payload, and a three-layer deterministic validator (schema, telecom semantics, live CAPIF registry) gates every call. Quality of Service (QoS) requests are authorized by the PCF of a containerized Open5GS 5G core with UERANSIM-emulated UEs. A live request takes 2.2–4.8 s with GPT-4o-mini or Claude Haiku 4.5 (6.9–8.2 s with Gemini), of which LLM planning accounts for 78–99%. On 300 unique bilingual intents (three runs each), the LLM planners reach an Intent-to-API Workflow Success Rate (IWSR) of 95.3–96.6%, compared with 83.3% for a keyword baseline and 36.0% for native function calling. Code, testbed and data: https://github.com/doadurak/mini-telco-platform.**

***Index Terms*—CAPIF, CAMARA, intent-based networking, LLM orchestration, 5G API exposure, Open5GS**

I. Introduction

Zero-touch Beyond-5G management requires orchestration layers that translate high-level intents into standards-compliant API invocations [1]. 3GPP CAPIF [2] and CAMARA [3], [4] standardize how verticals discover and consume network capabilities, but closing the intent-to-API gap on a *live* exposure stack remains open. OpenCAPIF [5] provides exposure infrastructure without NL planning; Turkovic et al. [6] ran live CAMARA QoD with hand-written payloads; Maatouk et al. [7] identify hallucination as the key risk of telecom LLMs, and Liu et al. [8] report 40% JSON-schema compliance for ungrounded GPT-4. Our open platform combines live CAPIF onboarding, schema-grounded multi-LLM planning, a three-layer validator, and CAMARA execution down to a 3GPP PCF.

II. Platform Architecture

Planning: A FastAPI backend exposes `POST /orchestrate` (Fig. 1). Phase 1 selects one of four CAMARA services (QoD v1.1, QoS Profiles, QoS Provisioning, Location Retrieval v0.5) from a schema-grounded catalog; Phase 2 generates the payload from the operation's OpenAPI schema, which is injected into the prompt [9]. Intent-level QoS classes (e.g. `QOS_E_STREAMING`) are mapped by target bit rate to the operator profiles the network enforces (`QOS_E/S/M/L`).

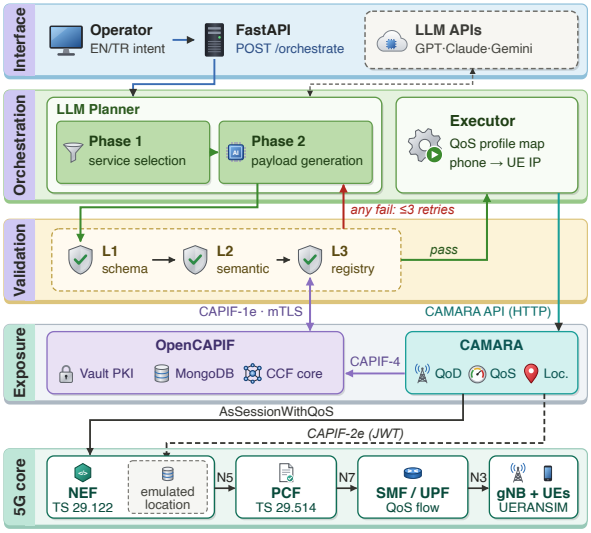


Fig. 1. Platform architecture. Any L1–L3 failure triggers at most three payload corrections; CAPIF-1e runs over mTLS, provider calls over HTTP.

Validation: L1 checks the payload against its Pydantic schema; L2 enforces telecom rules (E.164 numbers, QoS profiles that map to an operator profile, durations of 1–7200 s, unicast IPv4); L3 queries the live CAPIF registry to confirm that the selected service is published. L1–L3 cannot detect a registered but wrong service; therefore, a live plan whose service differs from the one chosen by the keyword planner is held until the operator confirms it.

Execution: For QoS, the adapter resolves the phone number to the UE's IP from the 5GC subscriber database; the CAMARA provider forwards an AsSessionWithQoS subscription to the NEF, which creates a PCF application session bound to the UE's PDU session. Location uses the NEF Monitoring Event API with emulated core data [10]. The CAMARA and NEF services are NCSRD's open-source implementations [11].

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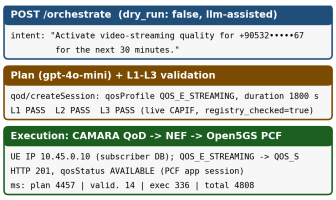


Fig. 2. Recorded live response (S1, GPT-4o-mini, `dry_run:false`, MSISDN masked).

TABLE I
IWSR AND HR (%), 300 INTENTS; LLMS: MEAN [RANGE], 3 RUNS

Planner	IWSR	Easy	Med.	Hard	HR
Keyword baseline	83.3	82.3	79.8	87.3	16.7
GPT-4o-mini	**96.6** [96.3, 96.7]	100	97.2	**93.0**	**3.0**
Claude Haiku 4.5	95.3 [95.3, 95.3]	100	100	87.3	4.0
Gemini 2.5 Flash	96.0 [95.7, 96.3]	99.3	99.6	90.0	3.2
GPT function calling	36.0	37.5	42.6	29.1	5.0
+ L1–L3 feedback	96.0	100	98.9	90.0	4.0

III. LIVE DEMONSTRATION

The full stack of Fig. 1, including Open5GS [12] and UERANSIM [13], runs on one 16 GB host via `start.sh`. All scenarios run with `dry_run:false`, and their responses are in the repository. **S1 (QoD):** *"Activate video-streaming quality for ⟨MSISDN⟩ for the next 30 minutes"* yields `QOS_E_STREAMING`, 1800 s, enforced as `QOS_S`; the PCF accepts the session and QoD returns HTTP 201, `AVAILABLE` (Fig. 2). **S2 (Location):** the NEF returns a WGS-84 area (HTTP 200). **S3 (ambiguity):** dataset entry *"Provide network quality improvement for device ⟨MSISDN⟩ for a long time"* (ground truth: QoD). All three LLMs select QoS Provisioning in every run, and the plan passes L1–L3; because the keyword planner selects QoD, the cross-check holds the call until it is confirmed, after which the PCF accepts it (`ACTIVE`). **S4 (backend switch):** after each `POST /config/provider` switch, S1 yields identical CAMARA payloads from all three LLMs. UERANSIM lacks NGAP PDU Session Resource Modify, so the dedicated QoS flow is authorized by the core but not enforced on the emulated radio.

IV. EVALUATION

The dataset holds 300 unique intents (152 EN / 148 TR; 96 easy, 94 medium, 110 hard): 275 across the four services and 25 ambiguous edge cases. We derived the ground truth (service, operation, payload checks) from the CAMARA OpenAPI constraints and NEF test fixtures; independent multi-annotator labelling, in particular of the 25 ambiguous intents, is left for future work. IWSR is the share of plans that select the correct service and pass validation; HR is the share of wrong service selections. Each LLM planner ran three times (temperature 0.1) without provider calls; every plan targets a published service, so L3 never rejects one. The keyword rules were written for a 60-intent pilot (100% there, 79.2% on the other 240; LLMs 95.4–96.5%). IWSR varies by at most 0.6 points across runs; the 10–14 failures per run are mostly QoD vs. QoS Provisioning confusions on hard intents (as in S3). Across

TABLE II
FAULT INJECTION INTO CORRECT PLANS (LIVE L3)

Layer	Injected fault	Caught	Repaired	Time (iter.)
L1	missing field, wrong type	40/40	40/40	1.3 s (1)
L2	E.164, duration, profile, IPv4	80/80	79/80	1.3 s (1)
L3	service not published	20/20	0/20	3.8 s (3)

the nine LLM runs, the cross-check holds 75 of these 92 wrong selections before execution, at the cost of a confirmation for 16% of correct plans. *Native function calling* (one tool per CAMARA operation, OpenAPI schema as parameters) selects services almost as well (HR 5.0%) but omits required fields that the intent does not state (e.g. `applicationServer`); our feedback loop repairs 181 such plans (2–3 per run for the two-phase planner). *Fault injection* (Table II): every fault is caught by its layer; GPT-4o-mini feedback repairs L1/L2 faults in one iteration but cannot fix an unpublished service. *Scalability and LLM APIs:* with live L3, one backend process serves ≈160 req/s with the keyword planner; with GPT-4o-mini, latency stays at ≈2.6 s up to 10 concurrent requests, so the LLM API is the bottleneck. If it fails, `auto` mode uses the keyword planner. As intents (with MSISDNs) reach an external API, production use needs pseudonymization or a self-hosted model. Finally, a demo video is available at https://youtu.be/RiT6QGwS9Pc.

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